

EMORY UNIVERSITY
Epilepsy & Clinical Neurophysiology

Setting Things Up:

BEFORE you plan to do the thalamic trial, follow the steps to set up the stimulation:

1. **Stop** Current Study (You can click the x on the RIGHT upper corner)
2. Click **“Review Study”** in the study list
3. Click **“Edit”**
4. **“Settings”**
5. Click **“Acquisition”** tab
6. Click **Advanced**
7. **Uncheck** 1) enable digital trigger, 2) use Nicolet cortical stim
8. OK -> Apply -> Save the changes
9. Return to the study

You will pick the target contacts on the Natus -> DSM control, but adjust parameters, and perform continuous stim on the Nicolet.

1. On Nicolet, change the duration to “continuous”
2. Use a timer to keep time (30 seconds, 20 mins, 1 hour)
3. IF you do not see stim artifact on EEG, click “OPEN/CLOSE relay”, or “START” on the DSM control

THALAMIC short trial

Recommended for: hippocampal onsets with abundant spiking; FCD with nearly continuous repetitive spiking

- Set stim (100 Hz, 100us)
- Start at 1mA and stimulate for one minute to test tolerability
- Increase by 1mA to reach 6mA, 30 seconds at a time
- Leave it running on continuous for 20 minutes (can decrease to 5-10 minutes for FCD with continuous spiking if the effect is immediately obvious)
- Turn it off for 5 minutes before trialing another contact pair

- If time allows, consider testing 5 Hz stimulation with 200 us pulse width up to 4mA (this could help determine patients that can be set to low frequency to avoid mood/cognition side effects)

THALAMIC prolonged trial

Recommended for: broad network onsets; quantitatively testing pre vs. post CCEPs

- Set stim at 100 Hz, 100us
- Start at 1mA and stimulate for one minute to test tolerability
- Increase by 1mA to reach 6mA, one minute at a time
- Leave it running on continuous for 120 minutes
- Turn it off for 15 minutes before trialing another contact pair

How to stim effect?

- Ideally, we will use a spike detector to quantify pre and post. In the absence of this pipeline for the time being, if the effect is not apparent, use the recommended workflow below:
- Determine your baseline interictal spiking rate (choose 3 x 10' baseline samples and count spikes per channel on main channels, average the spikes across all three epochs)
- Determine the spiking rate on the 10' epoch between the post-stim 5 min mark and before the start of the next trial (e.g. if LPUL1-2 was stimulated at 10:03 for 120 minutes and turned off at 12:03, ignore 12:04-12:09, and count spikes between 12:40 to 12:50).
- Repeat for each trial.